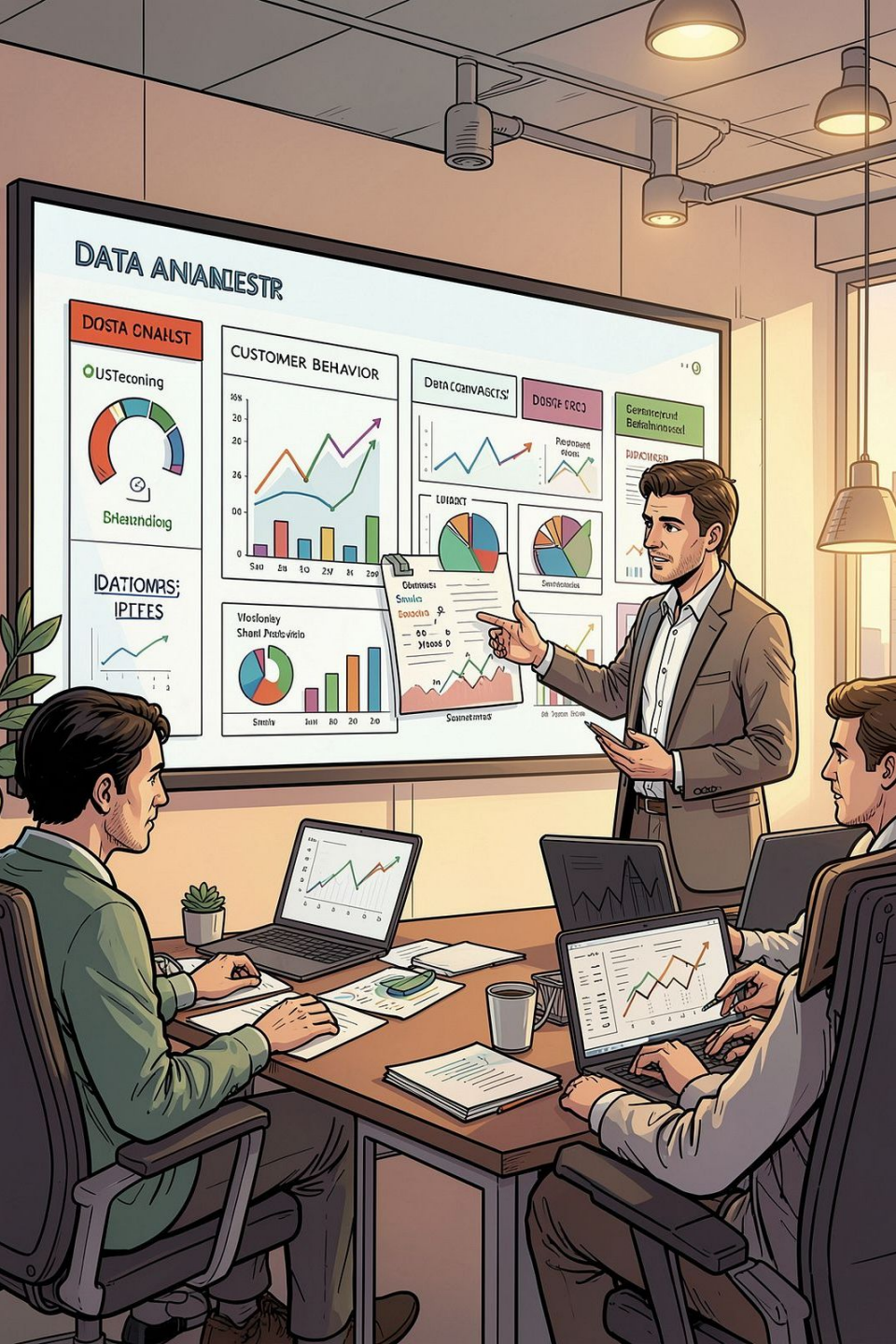
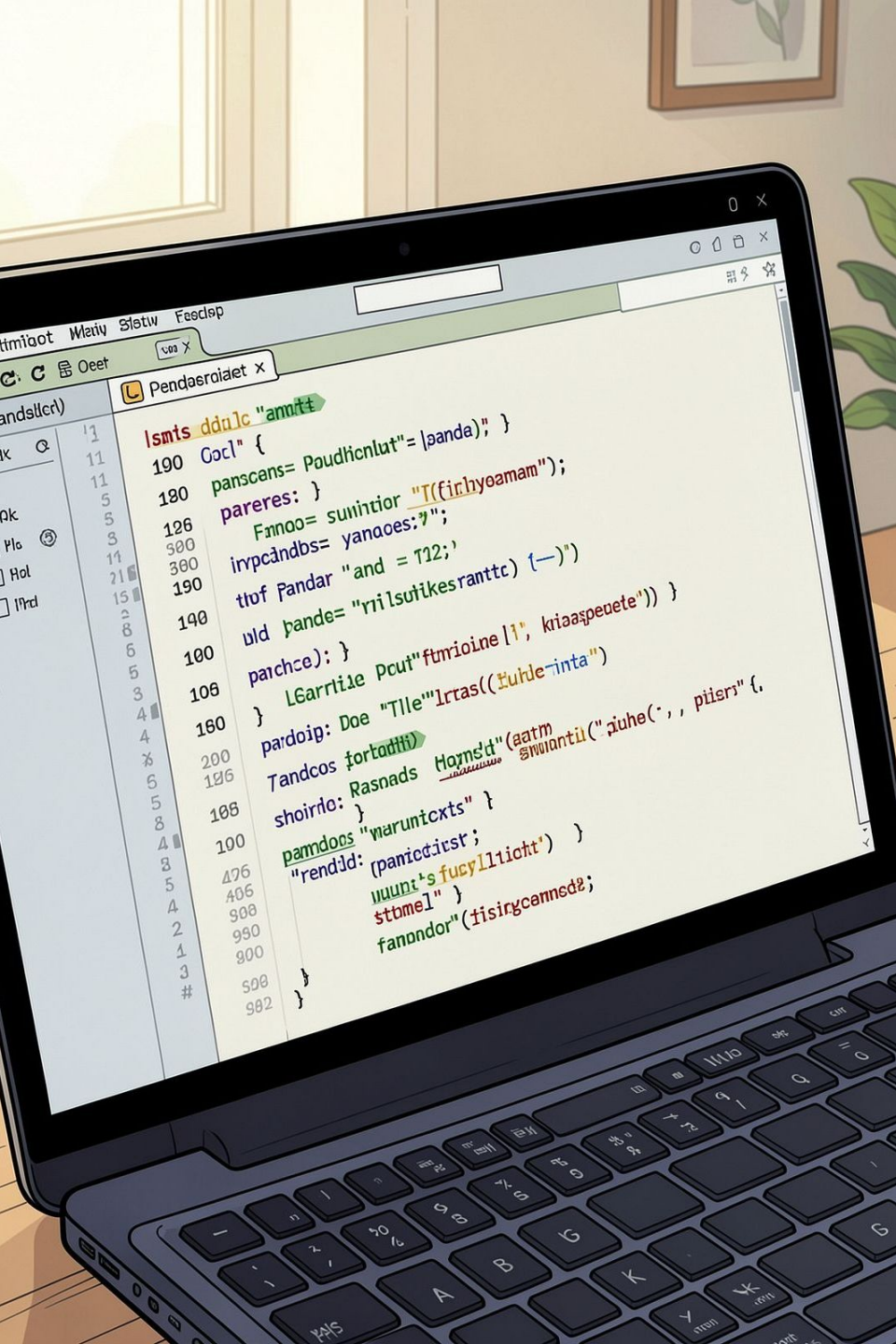




Customer Shopping Behavior Analysis

Analysis of 3,900 transactions to reveal spending patterns, segments, product preferences, and subscription behavior to guide strategy.



Dataset Snapshot

Scope

3,900 rows • 18 columns

Key Features

Demographics, purchase details,
shopping behavior, shipping,
discounts

Missing Data

37 nulls in Review Rating

Typical Purchase

Mean purchase amount: \$59.76 •
Avg rating: 3.75

Data Preparation & Feature Engineering



Loaded with pandas; used `df.info()` and `.describe()`. Imputed missing Review Rating by category median, standardized column names to snake_case, and engineered age_group and purchase_frequency_days. Dropped redundant promo_code_used and loaded cleaned data to MS_SQL.

SQL Analysis — High-level Findings

Revenue by Gender

Male: 157,890 • Female: 75,191

Shipping Comparison

Express avg: \$60.48
Standard avg: \$58.46

Subscribers vs Non-Subscribers

No (2847 customers) - Avg spend \$59.87
Yes (1053) - Avg spend \$59.49



High-Spending Discount Users

Identified customers who used discounts yet spent above average.
Several purchases show discount applied with amounts at or near \$100+ (multiple Customer_IDs flagged).



Top Products & Discount-Dependent Items



Top by Rating

Gloves (3.86)
Sandals (3.84)
Boots (3.82)
Hat (3.80)
Skirt (3.79)



Discount-Dependent

Hat (50%)
Sneakers (49.66%)
Coat (49.07%)
Sweater (48.17%)
Pants (47.37%)

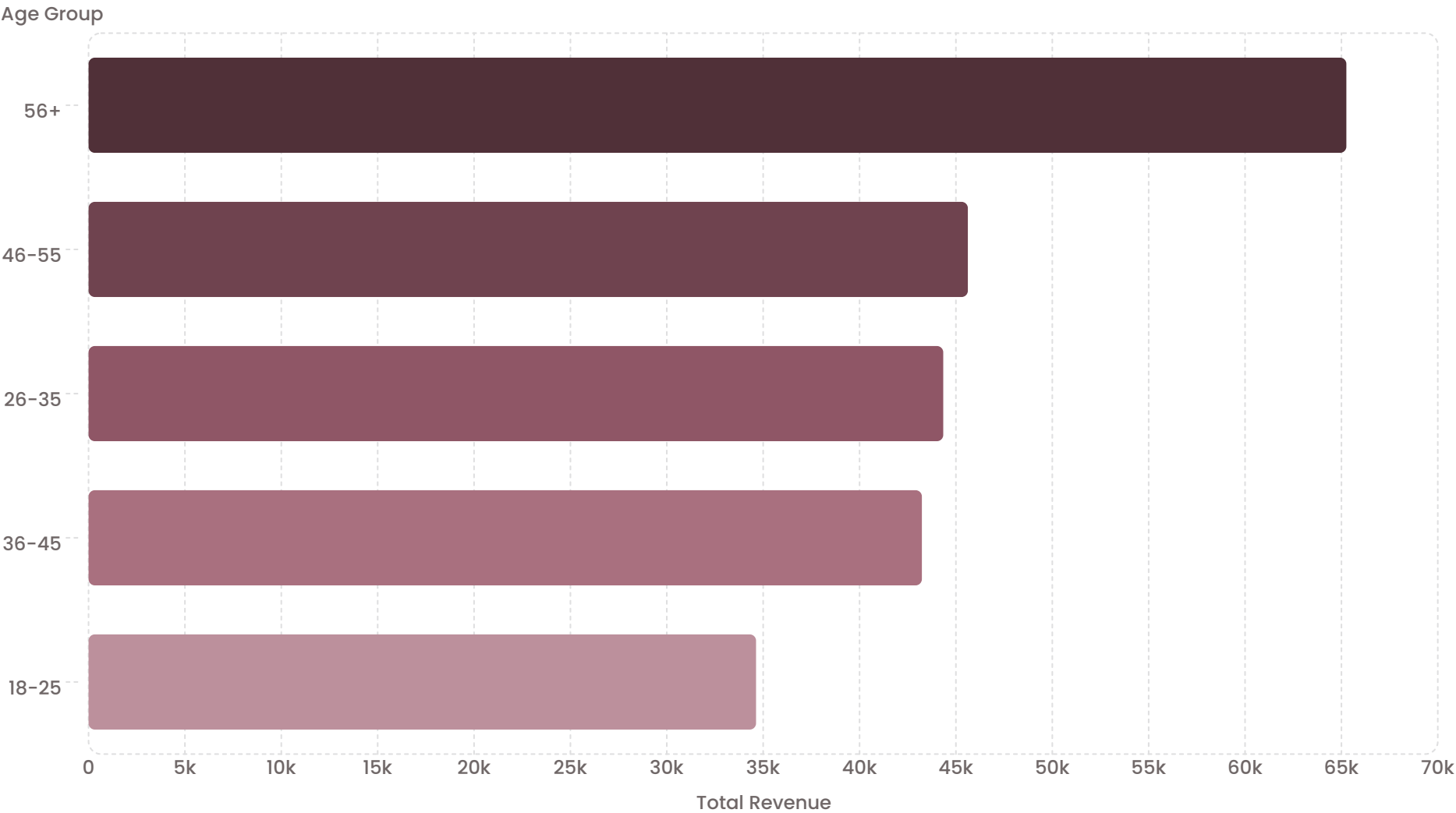
Customer Segmentation & Loyalty



Segments: Loyal 3,116 customers
Returning 701
New 83.

Recommendation: focus loyalty programs to retain high-value repeat buyers.

Revenue by Age Group



56+ contributes 28% of revenue; target high-revenue age groups for tailored campaigns.

Power BI Dashboard & Key Metrics



Core KPIs

3.9K customers • \$59.76 avg purchase • 3.75 avg rating

Subscription Split

Yes 27% • No 73%

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to raise subscriber share and lifetime value.

Customer Loyalty Programs

Reward repeat buyers to move them into the Loyal segment.

Review Discount Policy

Balance short-term sales with margin protection—focus on discount-dependent products.

Product Positioning & Targeting

Highlight top-rated best-sellers and target high-revenue age groups and express-shipping users.

- ❑ Next steps: A/B test subscription incentives and refine discount strategies based on margin impact.

